Geo-Tagging US Roadway Assets Using 2D Images

Project Team

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Chapter 1

Introduction

This document specifies the requirements for the development of a system aimed at geo-tagging US roadway assets using 2D images. The product leverages deep learning models to enhance the accuracy of asset geolocation, ensuring that assets such as traffic signs and other roadway infrastructure can be precisely mapped with a margin of error within 3 feet.

The document is intended for various types of readers, including:

- **Developers:** Responsible for the implementation of the system using the specified tools and frameworks such as Python, PyTorch, and OpenCV. Developers will utilize the requirements to design and code the functionalities of the system.
- **Project Managers:** Oversee the project timeline and ensure that development stays on track. They will use this document to monitor progress and ensure that the objectives are being met.
- **Marketing Staff:** Interested in understanding the product's capabilities and marketability, including its benefits for potential users like transportation agencies.
- **Users:** End users, including transportation agencies, who will directly interact with the system to retrieve geotagged asset data.
- **Testers:** Responsible for validating the accuracy of the geo-tagging system, ensuring that it meets the required maximum margin of error within 3 feet and works reliably under real-world conditions.
- **Documentation Writers:** Use this document to understand the technical specifications and user requirements to create manuals or guides for the system's use and maintenance.

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1.1 Problem Statement

In the rapidly evolving domain of transportation infrastructure, the accurate management of roadway assets is crucial for ensuring safety, efficiency, and long-term sustainability. Traditional methods of asset management, which often rely on manual inspections and 2D imagery, face significant challenges in precisely determining the geolocation of assets such as traffic signs, light poles, and other critical infrastructure elements. These challenges are worsened by the inherent limitations of 2D images, which lack depth information and are prone to inaccuracies due to camera angles, lighting conditions, and occlusions.

The primary problem addressed by this project is the significant margin of error in geolocation when using conventional methods to tag assets from 2D images. Existing approaches can lead to location errors that exceed acceptable thresholds, resulting in inefficiencies and potential safety hazards. Moreover, the reliance on manual processes is time-consuming, resource-intensive, and susceptible to human error.

This project aims to develop a software solution that leverages deep learning techniques to accurately geotag roadway assets within a 3-foot margin of error. By addressing the limitations of current methods, the proposed system will improve the precision of asset location identification, thereby enhancing the efficiency of infrastructure management and reducing the resources required for asset tracking and maintenance.

1.2 Scope

The scope of this project encompasses the development of a software system designed to accurately geotag roadway assets using 2D images. The primary focus is on creating a robust deep learning-based solution that can estimate the depth and geolocation of assets, such as traffic signs and light poles, within a 3-foot accuracy. The project includes the design, implementation, and validation of a model capable of processing 2D roadway images to predict the GPS coordinates of identified assets.

This system will integrate advanced monocular depth estimation models with geospatial data processing techniques to achieve high precision in geolocation. The functionalities covered within this project include image preprocessing, feature extraction, depth estimation, and geolocation prediction. The project will also involve the development of a user interface or API that allows transportation agencies to upload images and retrieve geotagged asset data.

The project will be developed using Python and relevant deep learning frameworks, such as TensorFlow or PyTorch. The scope is limited to processing static 2D images and does not include real-time video processing or integration with live camera feeds.

Finally, the project will deliver a working system, a comprehensive documentation covering the methodologies, algorithms, and technical implementation details. This documentation will ensure that future developers can understand and potentially build upon the work. The scope is carefully defined to ensure that the project is feasible within the given timeline and resources, while still providing a solution that is valuable for real-world applications.

1.3 Modules

The proposed project is divided into five key modules, each responsible for a critical component of the system's development. These modules work together to achieve the accurate geotagging of roadway assets using 2D images, from initial data preparation to final field testing and optimization. Below is a description of each module and its associated features.

1.3.1 Module 1: Preprocessing Dataset

This module is dedicated to enhancing the quality of the input images by addressing issues such as fogginess and blurriness through advanced preprocessing techniques. The objective is to ensure that the dataset used in subsequent stages is of the highest quality, which is critical for achieving accurate depth estimation and geolocation of roadway assets.

1. Implement image preprocessing methods to mitigate fog and blur, thereby improving overall image clarity.
2. Utilize specialized models for image enhancement, focusing on improving the visibility of assets in challenging visual conditions.

1.3.2 Module 2: Depth Map Model Selection

This module focuses on evaluating and selecting the most suitable depth estimation model for our specific application. Various state-of-the-art models will be rigorously tested to determine which one best meets the needs of our project. The chosen model will serve as the foundation for the subsequent stages of development.

1. Conduct a comprehensive evaluation of leading depth estimation models using the provided dataset.

2. Select the model that achieves the optimal balance between accuracy, performance, and computational efficiency.

1.3.3 Module 3: Fine-Tuning Depth Map Estimation Model

This module involves fine-tuning the selected depth estimation model on our specific dataset to ensure it accurately reflects the real-world conditions and asset types encountered in the project.

1. Fine-tuning the chosen depth estimation model using our dataset to improve its accuracy and reliability.
2. Adjusting model parameters and retraining to optimize performance for our specific application.

1.3.4 Module 4: Geolocation Estimation

This module is dedicated to calculating the geolocation of roadway assets using the depth maps generated in the previous module. It includes calculating distances and angles to accurately estimate the GPS coordinates of the assets.

1. Calculation of distance to assets using depth map data and angle measurements.
2. Estimation of GPS coordinates based on the calculated distances and angles.
3. Validation of the geolocation accuracy against ground truth data.

1.3.5 Module 5: Field Testing and Optimization

The final module focuses on field testing the entire system to validate its real-world performance and making necessary optimizations to ensure it meets the company's requirements.

1. Conducting field tests to evaluate the system's performance in real-world conditions.
2. Optimization of the model and system based on field test results to improve accuracy and efficiency.
3. Preparing the system for deployment, including final adjustments and documentation.

1.4 User Classes and Characteristics

This section identifies the different user classes and describes their pertinent characteristics.

User Class	Description
Transportation Agency Staff	The primary users of the system, responsible for managing roadway assets such as traffic signs and other infrastructure. They require an intuitive interface to upload images and retrieve GPS coordinates for asset locations. Although they may not have technical expertise in machine learning or computer vision, their focus is on data accuracy and efficiency for asset management tasks.
Developers and Data Scientists	These users will maintain, upgrade, and extend the system. They are experienced in deep learning, computer vision, and geospatial data processing, and they require access to the back-end of the system. They will need documentation on the architecture and algorithms used, and the system should be modular and extensible to allow for updates.
System Administrators	Responsible for managing the deployment and ongoing maintenance of the system, these users will handle tasks such as server setup, security management, and performance monitoring. They need a system that is easy to deploy on cloud or local infrastructure, with detailed logs and error handling mechanisms for troubleshooting.
Supervisors and Project Managers	Overseeing the development and implementation of the project, these users will focus on the high-level progress and performance of the system. They are interested in performance reports, progress dashboards, and geolocation accuracy, but they do not need access to the technical aspects of the system.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter includes the use case diagram which mentions all the use case scenarios of this project. Moreover this chapter demonstrates the functional and non functional requirements of the project. These requirements are essential for system which ensures if the system meets the needs of its users and operates efficiently.

2.1 Use-case

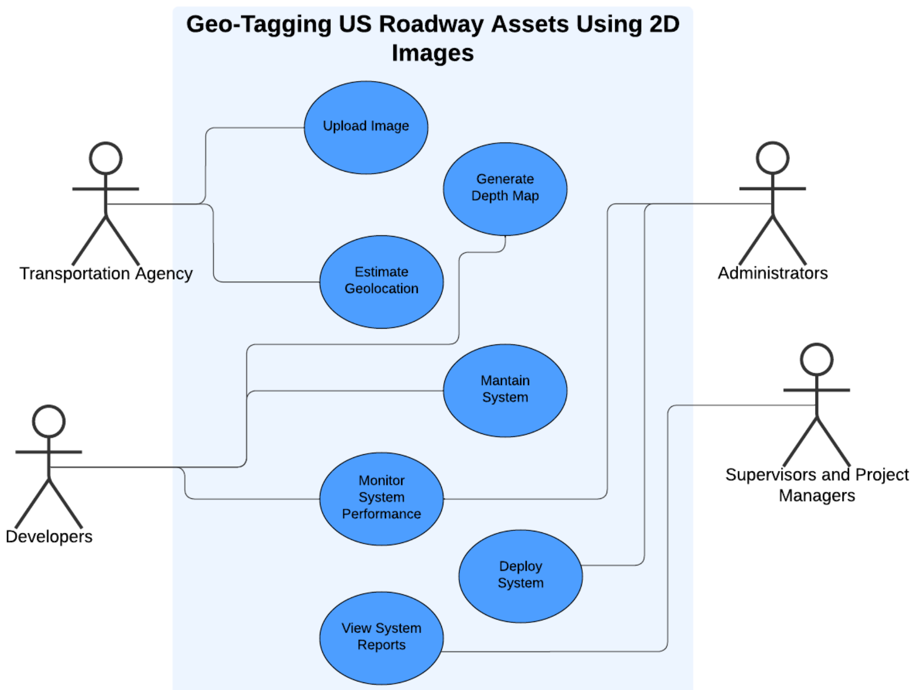

Figure 2.1: Use Case Diagram

Functional Requirements

Feature 1: Image Preprocessing

The system shall preprocess the dataset of road images to improve their quality for further analysis.

- FR-1 The system shall detect blurry images and apply deblurring techniques.
- FR-2 The system shall identify foggy images and apply defogging algorithms.
- FR-3 The system shall detect rainy images and remove water streaks using preprocessing techniques.

Feature 2: Depth Map Creation

The system shall estimate the depth of objects in the 2D images.

- FR-4 The system shall generate a depth map for each preprocessed image using a depth estimation model.
- FR-5 The system shall ensure that the depth map corresponds to the real-world dimensions of the scene in the image.

Feature 3: Road Sign and Bounding Box Metadata

The system shall utilize the metadata containing bounding box of road signs in the image.

- FR-6 The system shall use metadata associated with the image to identify the already detected road signs within the image.
- FR-7 The system shall extract the pixel values of the bounding box containing the road sign from the depth map.

Feature 4: Distance Calculation

The system shall calculate the real-life distance between the road sign and the camera.

- FR-8 The system shall convert the pixel values of the bounding box in the depth map to real-life units.

FR-9 The system shall calculate the distance between the road sign and the camera based on the converted depth values.

Feature 5: Geo-Coordinates Calculation

The system shall calculate the geolocation of the road sign.

FR-10 The system shall use the geolocation of the camera from the metadata to calculate the geo-coordinates of the road sign.

FR-11 The system shall calculate the angle of the road sign relative to the camera's position.

FR-12 The system shall use geo-calculation algorithms to determine the road sign's exact latitude and longitude.

Feature 6: Geo-Tagging and Display

The system shall geo-tag the detected road sign and display it on a map.

FR-13 The system shall attach the geo-coordinates to the detected road sign as metadata.

FR-14 The system shall display the geo-tagged road sign on a map using a Maps API.

FR-15 The system shall allow zooming and panning functionality for the map display.

Non-Functional Requirements

This section outlines the non-functional requirements for the system responsible for geo-tagging roadway assets using 2D images. The non-functional requirements describe the quality attributes, performance, and constraints that the system must meet to ensure its efficiency, reliability, and user satisfaction.

Performance Requirements

The system shall meet the following performance benchmarks:

NFR-1 The system shall process each image and produce geo-tagged results in under 5 seconds on average for images of 1920x1080 resolution.

NFR-2 The system shall be able to process a dataset of at least 10,000 images within 8 hours.

NFR-3 The depth estimation model shall operate with a latency of less than 2 seconds per image.

Scalability Requirements

The system shall be designed to scale for larger datasets and higher user traffic:

NFR-4 The system shall support scaling to process up to 30,000 images in one batch without a significant increase in processing time.

NFR-5 The system architecture shall allow for horizontal scaling by adding additional processing nodes to handle increased workloads.

Reliability Requirements

The system shall exhibit high reliability to ensure uninterrupted operation:

NFR-6 The system shall have an uptime of 99.9% or higher, with planned maintenance windows scheduled outside peak usage hours.

NFR-7 The system shall implement retry mechanisms for image processing in case of failure, ensuring that no image is left unprocessed.

NFR-8 The system shall be capable of recovering from failures within 2 minutes without loss of data or image processing progress.

Accuracy Requirements

The system shall ensure the accuracy of image processing and geotagging:

NFR-9 The system shall calculate geo-coordinates of road signs with a margin of error of no more than 3 feet.

Usability Requirements

The system shall ensure a high degree of usability for its users:

NFR-10 The user interface for the map display shall be intuitive and require no more than 30 minutes of training for new users.

NFR-11 The system shall allow users to easily export geo-tagged results and reports without needing technical expertise.

NFR-12 The system shall provide clear and actionable error messages in case of any failures, guiding users toward resolution.

Security Requirements

The system shall implement strong security mechanisms to protect data:

NFR-13 The system shall use encryption (e.g., TLS) for all communications involving image data and geo-tagged results.

NFR-14 The system shall require user authentication via username and password before accessing any processing features.

NFR-15 The system shall log all user actions and changes to the geotagging data for auditing purposes.

Maintainability Requirements

The system shall be easy to maintain and update:

NFR-16 The system codebase shall follow standard coding practices, ensuring that new developers can understand the codebase within 2 weeks.

NFR-17 The system shall allow for updates and feature additions with minimal downtime, using a continuous integration/continuous deployment (CI/CD) pipeline.

Compatibility Requirements

The system shall be compatible with various external tools and APIs:

NFR-18 The system shall integrate with multiple map APIs (e.g., Google Maps, OpenStreetMap) for displaying geo-tagged signs.

NFR-19 The system shall support common image formats such as JPEG, PNG, and TIFF for input images.

NFR-20 The system shall be compatible with both Windows and Linux operating systems for installation and operation.

Legal and Regulatory Requirements

The system shall comply with relevant legal and regulatory standards:

NFR-21 The system shall comply with data protection regulations (e.g., GDPR) to ensure that no sensitive data is improperly exposed or misused.

NFR-22 The system shall obtain explicit user consent before collecting or processing geolocation data.

Chapter 3

System Overview

This chapter provides an overview of the system, detailing its architecture, design models, and data design. The project involves developing a system for geo-tagging roadway assets using 2D images, leveraging machine learning, image processing, depth estimation, and geolocation algorithms to accurately estimate GPS coordinates within a margin of error of 3 feet.

3.1 Architectural Design

The system is organized into several modules that collaborate to achieve the geo-tagging of roadway assets: image preprocessing, depth map generation, geolocation estimation, and field testing.

The architectural design is multi-tiered, with each module responsible for a specific task within the overall workflow.

Description:
The high-level architectural diagram provides an overview of the system components. It illustrates how images captured by the surveillance vehicle are processed, how depth estimation is performed, and how geolocation is determined. Each module plays a crucial role in ensuring the system's accuracy and reliability.

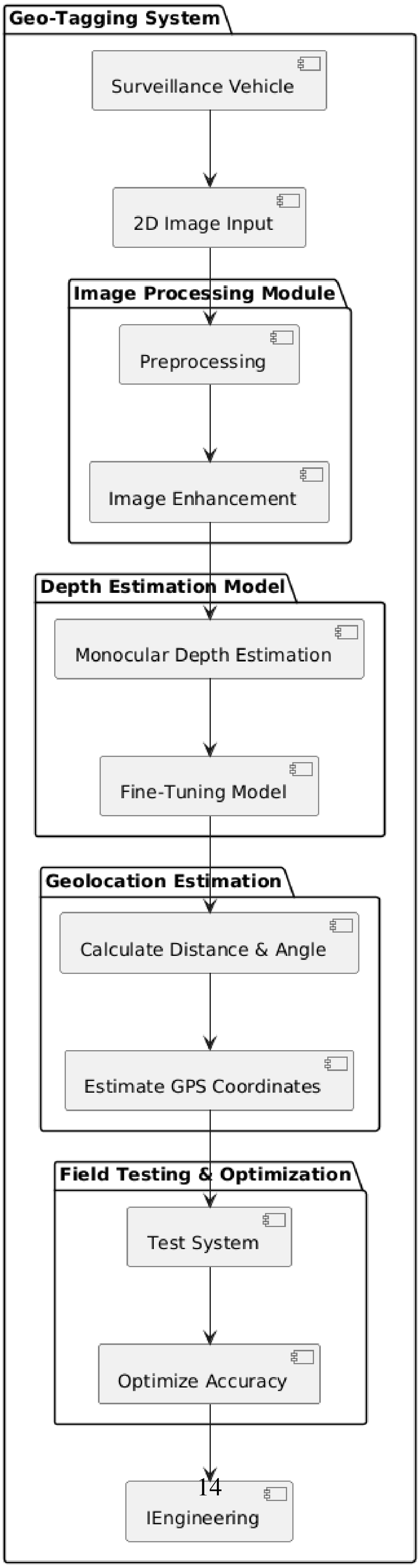

Figure 3.1: High-Level Architecture of the Geo-Tagging System

3.2 Design Models

This section presents various design models that represent different aspects of the system, including process flow, data flow, system interactions, and data representation.

3.2.1 Process Flow (Activity Diagram)

The process flow diagram describes the sequence of actions the system performs, from capturing images to field testing.

Description:
The diagram illustrates the activity flow, beginning with image preprocessing proceeding through depth estimation and geolocation estimation, and coming to an end with field testing and system optimization.

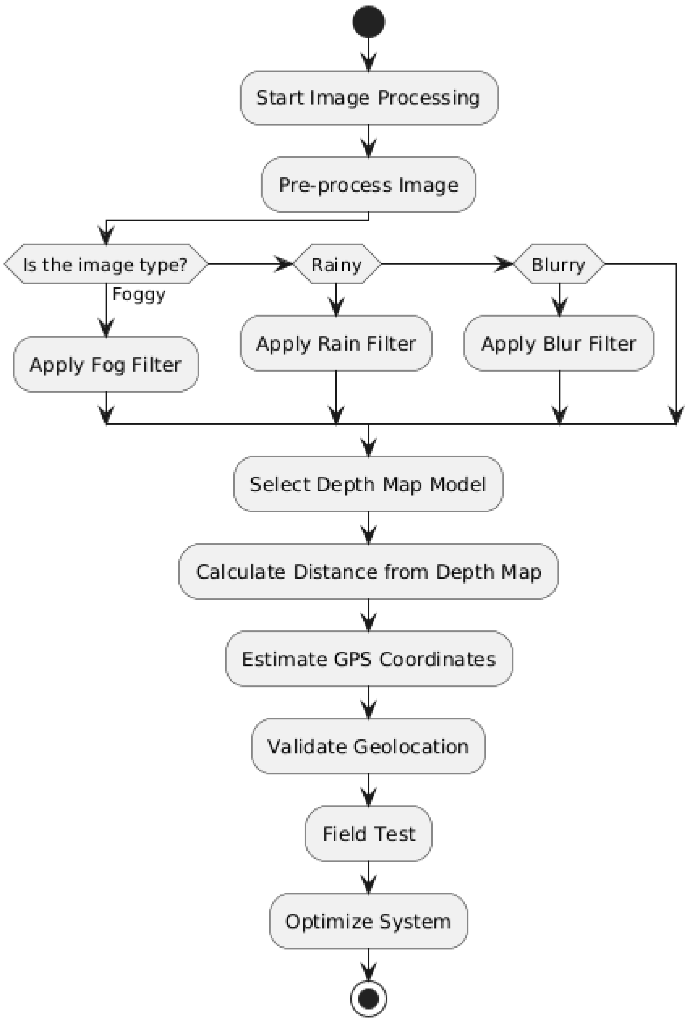

Figure 3.2: Process Flow (Activity Diagram)

3.2.2 Data Flow Diagram

The data flow diagram represents the movement of data between system components. It shows how images are processed, depth maps are generated, and GPS coordinates are calculated.

Description:
The diagram illustrates how data is transferred between different modules of the system, from image capture to the final GPS estimation.

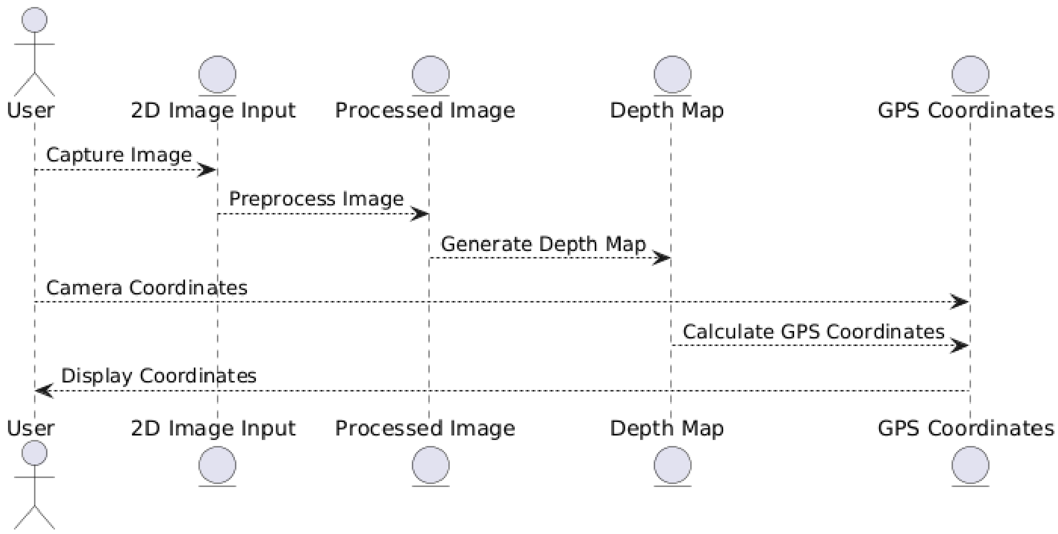

Figure 3.3: Data Flow Diagram

3.2.3 System-Level Sequence Diagram

The system-level sequence diagram details the interactions between the main components of the system, focusing on the workflow from capturing an image to displaying the final GPS coordinates to the user.

Description:
This diagram shows the sequence of operations performed by system components, including image processing, depth estimation, and geolocation estimation.

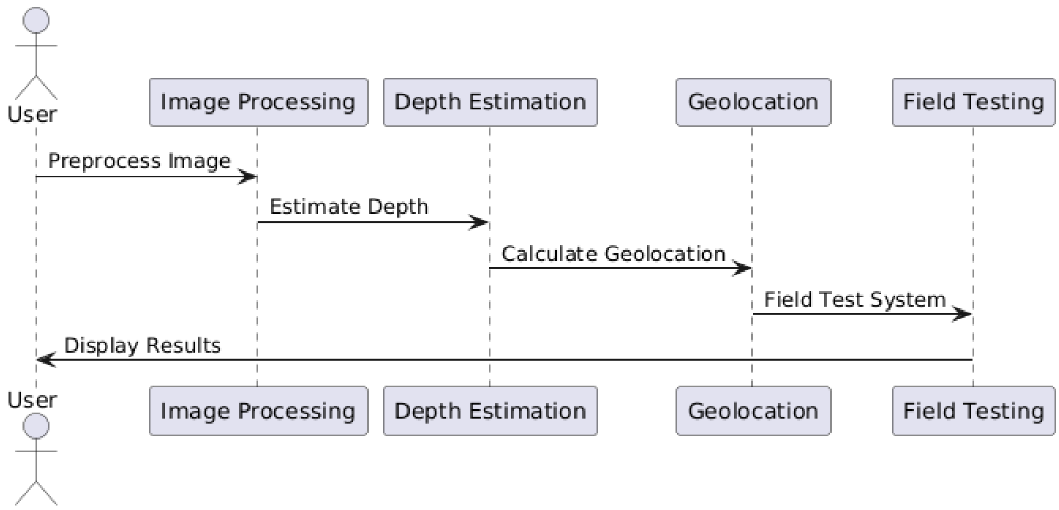

Figure 3.4: System-Level Sequence Diagram

3.2.4 Data Representation Diagram (ERD)

An Entity-Relationship Diagram (ERD) has been selected to represent the system's data structure. The ERD models the relationships between key entities such as images, assets, bounding boxes, GPS data, and depth maps.

Description:
This diagram shows the relationships between various entities, including 'Image', 'BoundingBox', 'GPSData', and 'DepthMap'. The ERD provides a detailed view of how these entities are interconnected and how data flows between them.

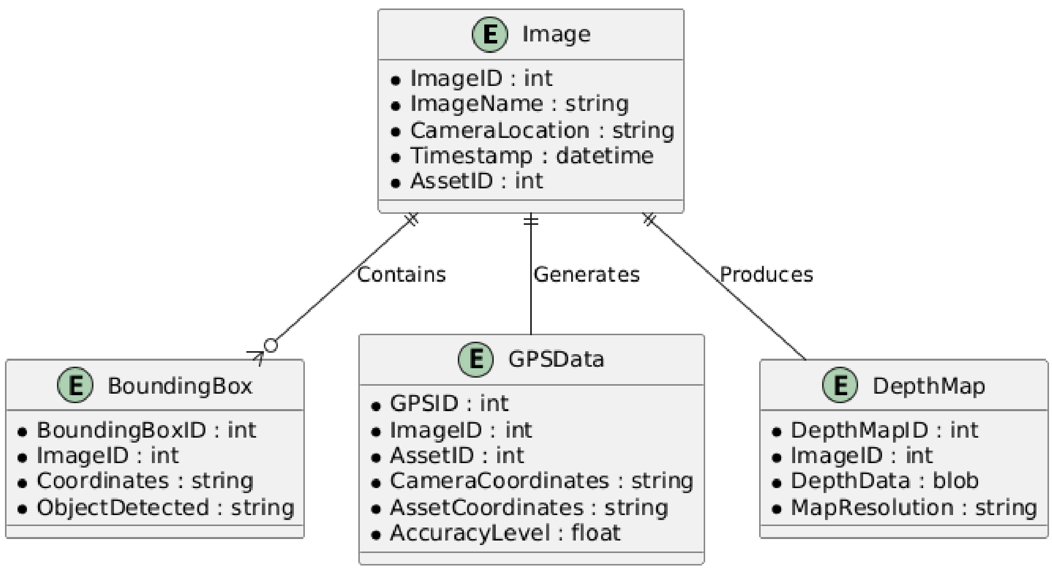

Figure 3.5: Data Representation (Entity-Relationship Diagram)

3.3 Data Design

The information domain of the system comprises 2D images captured by surveillance vehicles, depth maps generated from those images, and GPS data calculated based on the depth information. The major entities stored in the system include ‘Image‘, ‘BoundingBox‘, ‘GPSData‘, and ‘DepthMap‘. These entities are linked in a structured format to ensure accurate geolocation estimation.

Databases or Data Storage:
The system utilizes a relational database to store data related to images, and GPS coordinates, as outlined in the ERD provided above. Each image is associated with an asset and its GPS coordinates, while bounding boxes and depth maps are used for finer details of object detection.

Data Processing:
Data processing in the system involves several stages: image preprocessing, feature extraction, depth map generation, GPS coordinate calculation based on depth data, and the application of error correction algorithms to ensure high accuracy.

Bibliography

[1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.